

Practice program:

1. Palindrome number:

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Q1 Palindrome number/string 10 marks

Palindrome number/string

Your Code

```
1 public class main{
2     public static void main(String[] args){
3         int n=121;
4         int rev=0;
5         int temp=n;
6         while (temp!=0){
7             rev=rev*10+temp%10;
8             temp=temp/10;
9         }
10        if(n==rev)
11            System.out.println("palindrome");
12        else
13            System.out.println("not palindrome");
14    }
15 }
```

Input

121

Output

palindrome

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2. First n Perfect numbers:

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Q2 First n perfect numbers 10 marks

First n perfect numbers

Your Code

```
1 public class main{
2     public static void main(String[] args){
3         int n=2;
4         int count=0;
5         int num=1;
6         while(count<n){
7             int sum=0;
8             for(int i=1;i<num;i++){
9                 if(num%i==0)
10                    sum=sum+i;
11             }
12             if(sum==num){
13                 System.out.println(num);
14                 count++;
15             }
16             num++;
17         }
18     }
19 }
```

Input

stdin input

Output

6
28

3.Decimal Number equivalent to binary number and octal number

Q3 Decimal number equivalent to binary number and octal number10 marks

Decimal number equivalent to binary number and octal number

Your Code

```
1 public class main{
2     public static void main(String[] args){
3         int binary =1011;
4         int decimal=0,base=1;
5         while(binary>0){
6             int rem=binary%10;
7             decimal+=rem*base;
8             base*=2;
9             binary/=10;
10        }
11        System.out.println(decimal);
12    }
13 }
```

RunSave

Input

stdin input

Output

11

4.Nth max number and nth min number in array then sum of it and difference of it

Q4 Nth max number and nth min number in array then sum of it and difference of it10 marks

Nth max number and nth min number in array then sum of it and difference of it

Your Code

```
1 import java.util.Arrays;
2
3 public class NthMaxMin {
4     public static void main(String[] args) {
5         int[] a = {8, 3, 10, 5, 1, 6};
6         int n = 2;
7         Arrays.sort(a);
8         int nthMin = a[n - 1];
9         int nthMax = a[a.length - n];
10        System.out.println("Nth Min = " + nthMin);
11        System.out.println("Nth Max = " + nthMax);
12        System.out.println("Sum = " + (nthMin + nthMax));
13        System.out.println("Difference = " + (nthMax - nthMin));
14    }
15 }
```

RunSave

Input

stdin input

Output

Nth Min = 3
Nth Max = 8
Sum = 11
Difference = 5

5.Find the average positive and negative number upto -1

Q5 Find the average positive and negative numbers upto -1

10 marks

Find the average positive and negative numbers upto -1

Your Code

```
1 public class Average {
2     public static void main(String[] args) {
3         int[] a = {10, 20, -5, 15, -10, -1};
4         int ps = 0, ns = 0, pc = 0, nc = 0;
5         for (int i = 0; a[i] != -1; i++) {
6             if (a[i] > 0) {
7                 ps += a[i];
8                 pc++;
9             } else {
10                ns += a[i];
11                nc++;
12            }
13        }
14
15        System.out.println("Positive Average = " + (ps / pc));
16        System.out.println("Negative Average = " + (ns / nc));
17    }
18 }
```

Input

stdin input

Output

```
Positive Average = 15
Negative Average = -7
```

6.Removing duplicates in array

Q6 Removing duplicates in array

10 marks

Removing duplicates in array

Your Code

```
1 public class RemoveDuplicates {
2     public static void main(String[] args) {
3         int[] a = {1, 2, 3, 2, 4, 1, 5};
4         for (int i = 0; i < a.length; i++) {
5             int j;
6             for (j = 0; j < i; j++)
7                 if (a[i] == a[j])
8                     break;
9             if (i == j)
10                System.out.print(a[i] + " ");
11        }
12    }
13 }
```

Input

stdin input

Output

```
1 2 3 4 5
```

7.Matrix Multiplication

Q7 Matrix multiplication

Matrix multiplication

Your Code

```
1 public class name{
2     public static void main(String[] args){
3         int[][] a={{1,2},{3,4}};
4         int[][] b={{5,6},{7,8}};
5         int[][] c=new int[2][2];
6         for (int i=0;i<2;i++)
7             for (int j=0;j<2;j++)
8                 c[i][j]=a[i][0]*b[0][j]+a[i][1]*b[1][j];
9         for (int i=0;i<2;i++){
10             for(int j=0;j<2;j++)
11                 System.out.print(c[i][j]+" ");
12             System.out.println();
13         }
14     }
15 }
```

Input

stdin input

Output

```
19 22
43 50
```